

Matcap Maker Usage Guide

Matcap Maker is a tool for creating Matcap (Material Capture) textures for 3D models. By layering multiple effects (lights, fresnel, noise, images, etc.), you can intuitively create original Matcaps.

Language Support

This application supports both English and Japanese. You can switch languages from the "Options" menu. Please note that changing the language requires restarting the application.

Interface Overview

The application is divided into three main areas:

1. Preview Window (Center)

- Displays a real-time rendering of your Matcap on a sphere.
- Use the "Export Image" button at the bottom to save the current result as an image file (PNG/JPG).

2. Layer List (Left)

- Lists all layers currently in the project.
- Processing occurs from top to bottom.
- **Add:** Create a new layer using the "Add Layer" button.
- **Remove:** Delete the currently selected layer.
- **Toggle Visibility:** Click the circular button next to each item to enable/disable the layer.
- **Reorder:** Drag and drop layers to change their order.
- **Context Menu:** Right-click on a layer to access "Duplicate" or "Delete" options.

3. Properties Panel (Right)

- Adjust detailed settings for the layer selected in the list.

Layer Types and Properties

1. Base Layer

Sets the foundation color and preview settings for the Matcap.

- **Color:** Sets the base color.
- **Preview Options:**
 - **Mode:**
 - **Standard:** Standard sphere preview.
 - **With Normal Map:** Previews how the Matcap looks with a normal map applied.
 - **Normal Map:** Select a normal map image to load.
 - **Strength/Scale/Offset:** Adjust the application of the normal map.

2. Spot Light

Adds lighting effects from a specific direction. Ideal for creating highlights.

- **Blend Mode:** Select how the light blends (Add, Screen, Overlay, etc.).

- **Intensity:** The brightness of the light.
- **Range:** The spread of the light.
- **Blur:** The softness of the light edges.
- **Scale X/Y:** Changes the aspect ratio of the light shape.
- **Rotation:** Rotates the light shape.
- **Direction X/Y/Z:** Sets the direction the light is coming from.
- **Color:** The color of the light.

3. Fresnel / Rim

Adds light to the contours (edges) of the object. Used for rim lights and material definition.

- **Intensity:** The brightness of the light.
- **Power:** The sharpness of the rim. Higher values push the light closer to the edge.
- **Bias:** Offsets the range of the effect.
- **Color:** The color of the light.

4. Noise

Adds overall grain or texture.

- **Intensity:** The strength of the noise.
- **Scale:** The fineness of the noise grain.
- **Seed Offset:** Changes the random seed for the noise pattern.
- **Color:** The color of the noise. Usually used with "Multiply" blend mode for shading.

5. Image

Maps an arbitrary image as a texture. Used for reflections or patterns.

- **Image:** Select the image file to use.
- **Mapping:**
 - **UV:** Maps the image according to the sphere's UV coordinates.
 - **Planar:** Maps the image like a planar projection.
- **Offset X/Y:** Adjusts the position of the image.
- **Scale:** Adjusts the size of the image.
- **Rotation:** Adjusts the angle of the image.
- **Blur:** Blurs the image.
- **Opacity:** Adjusts the transparency of the image.

6. Color Adjustment

Adjusts the overall tone of the composite result up to this layer.

- **Hue:** Adjusts the hue.
- **Saturation:** Adjusts the color intensity.
- **Brightness:** Adjusts the brightness.
- **Contrast:** Adjusts the contrast.

7. Gradient

Renders a smooth color gradient across the sphere surface. Can be combined with any blend mode for effects such as ambient color transitions (sky/ground), directional tonal variation, and color overlays.

- **Blend Mode:** Select how the gradient blends with layers below (Normal, Multiply, Add, etc.).
- **Angle:** The direction of the gradient in degrees (0–360°).
 - 0° = left → right, 90° = bottom → top, 270° = top → bottom.
- **Type:** Selects the gradient style.
 - **Linear:** A straight gradient in the direction specified by Angle.
 - **Radial:** A gradient radiating outward from the center of the sphere.
- **Gradient Colors:** Visual gradient editor with up to 8 color stops.
 - **Click the gradient bar** to add a new color stop at that position (initial color is sampled from the existing gradient).
 - **Drag a pin** left or right to adjust its position.
 - **Click a pin** to select it — edit its color and exact position using the controls below.
 - **Double-click a pin** to open the color picker directly.
 - **Position spinbox:** Fine-tune the selected stop's position (0.0–1.0).
 - **Delete button:** Remove the selected stop (requires at least 2 stops to remain).

8. Blur / Sharpen

A post-process layer that applies a blur or sharpen to the entire composite result up to this layer. Place it above the layers you want to affect.

- **Blend Mode / Opacity:** Control how the effect is blended back in. Opacity also acts as the overall strength of the effect.
- **Mode:**
 - **Blur:** Softens the image using a high-quality separable Gaussian blur. The Matcap circle keeps its original size — only the content inside it is blurred (the edges are not expanded outward).
 - **Sharpen:** Emphasizes edges and detail using an unsharp mask.
- **Radius:** The size of the blur (also the internal radius used for sharpening). Larger values blur over a wider area.
- **Amount:** The strength of the sharpening (used in **Sharpen** mode). In **Blur** mode, use Opacity to control the strength.

Menu Operations

File

- **Open Project:** Opens a saved project file (.json).
- **Save Project:** Saves the current state as a project. Used image assets are also copied to an **assets** folder.
- **Export Image:** Exports the Matcap image, same as the button at the bottom.

Options

- **Language:** Switches language between English and Japanese (Release requires restart).
- **Resolution:** Selects the export resolution (64x64 to 4096x4096).
- **Padding:** Sets the margin for the exported image.

Help

- **Third Party Notices:** Displays legal notices for libraries used.